Ex no:-4 Setup and configuration of Local ...
using a switch and Ethernet cables.

Aim:
To setup and configure a local Area Network using a switch and Ethernet cables, assign IP address and verify communication between computer.

LAN:
A Local Area Network connects computers and devices within a small area such as a home, office, or school. It enables users to shares files, ~~or school, It enables users to shares files~~ printers, internet access, and other resource. A switch connects all device and allows fast and secure communication.

* Step to set up a LAN:

1) Plan and Design a appropriate network topology taking into account Network requirment and equipment Location.

2) Take 4 computers a switch with 8, 16 or 24 ports which is sufficient for Networks of these sizes and 4 Ethernet cables.

3) Connect the computer to Network switch via an Ethernet cables, which is a simple as plugging one end of the Ethernet cables into the computer and other end into the Network switch.

4) Assign IP address to your PC's
i) Log on the computer as Administration or as owner.
ii) Click Network and Internet Connection.
iii) Right click local Area connection / Ethernet
-> Go to properties -> Select Internet Protocol - Click on properties select use the following ip address option and assign ip address.

Similary assign Ip address to all the PCs connected to switch,

PC1 - IP address : 10.1.1.1, Subnet mask 255.0.0.0
PC2 - IP address : 10.1.1.2, Subnet mask 233.0.0.0
PC3 - IP address - 10.1.1.3, Subnet mask 233.0.0.0
PC4 - IP address - 10.1.1.4, Subnet mask 233.0.0.0

5) Configure the switch with the IP address 10.1.1.B

6) Check the connectivity between switch and other machine by using ping command in the command prompt to the device.

7) Select a folder → go to properties → click sharing tab → share it with everyone on the same LAN.

8) Try to access the shared folder from other computer of the networks.

Result:

A LAN was successfully configured using a switch and Ethernet cables. IP addresses were assigned correctly, communication was verify using the ping command and file sharing was successfully established between computers on the LAN